CS341 · COMPUTER GRAPHICS

Midterm 472 Solutions

20 marks

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01 Solution Q1: short answers

1. Why use aspect ratio?

Aspect ratios are useful for maintaining geometric features of figures drawn on a screen: a square remains a square and does not become a rectangle, and a circle stays a circle and does not become an ellipse.

2. Absolute vs relative coordinates

Absolute: the values are the actual positions within the coordinate system in use (measured from the origin).
Relative: the position is given as an offset from the last referenced position (the current position).
Example: current position (3, 8) + relative (2, −1) → absolute (5, 7).

3. Define a vector

A vector is a quantity that has **magnitude and direction** (like velocity). It represents a displacement and has no fixed location.

4. Vector from P to Q

```
v = Q − P
```

5. Rays emitted by a cathode ray tube

The rays are **electrons** (an electron beam). When they hit the phosphor coating of the screen, the phosphor emits **light**.

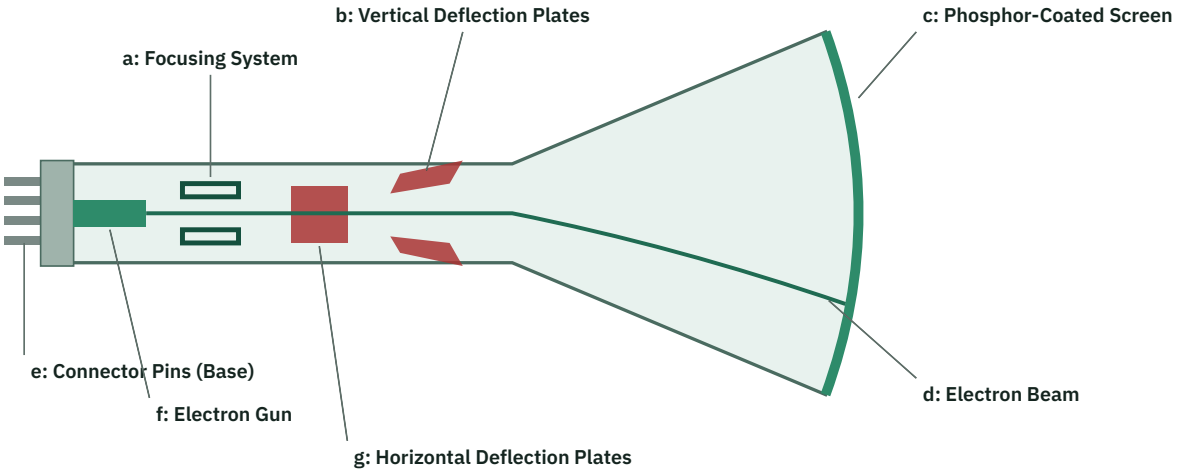
02 Solution Q2: true / false

#	Answer	Reason
1	**True**	This is the definition of a vector.
2	**False**	The sum of two points has 2 as its last homogeneous coordinate; a valid point needs coefficients summing to 1 (an affine combination).
3	**True**	The angle is 90° and `cos 90° = 0`, so the dot product is 0.
4	**False**	A unit vector has length exactly 1.

03 Solution Q3: multiple choice

#	Answer	Reason
1	**B) Frame buffer**	In raster systems the picture is stored in the refresh (frame) buffer.
2	**B) Random scan**	In random scan the electron beam directly draws the picture; raster sweeps the screen line by line.
3	**C) Dot pitch**	Clarity depends on resolution, dot pitch and refresh rate.
4	**C) Mouse**	The others are output devices.
5	**D) Data encryption**	The three tasks are modeling, rendering and animation.
6	**D) All of the above**	All three are applications on the slides.
7	**A**	Creation, storage and manipulation of models and images.
8	**A**	The same definition.

04 Solution Q4: CRT labels



Letter	Part
a	Focusing System
b	Vertical Deflection Plates
c	Phosphor-Coated Screen
d	Electron Beam
e	Connector Pins (Base)
f	Electron Gun
g	Horizontal Deflection Plates

05 Solution Q5: DDA

$dx = 4 - 2 = 2$ $\quad\quad$ $dy = 10 - 2 = 8$

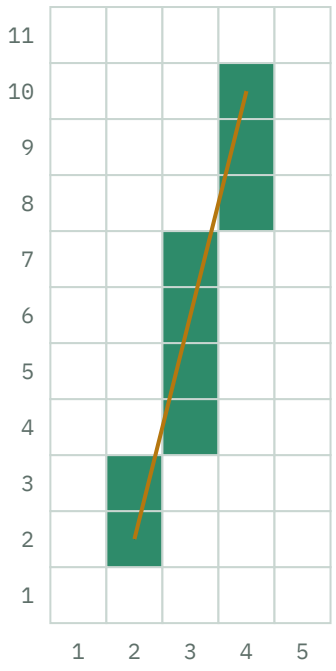
$m = \frac{8}{2} = 4 \rightarrow |m| > 1 \rightarrow$ step in y

$steps = max(|dx|, |dy|) = 8$

$xInc = \frac{2}{8} = 0.25$ $\quad\quad$ $yInc = \frac{8}{8} = 1$

$round(a) = int(a + 0.5)$

k	x	y	round(x)	round(y)
0	2	2	2	2
1	2.25	3	2	3
2	2.5	4	3	4
3	2.75	5	3	5
4	3	6	3	6
5	3.25	7	3	7
6	3.5	8	4	8
7	3.75	9	4	9
8	4	10	4	10



Points: (2,2) (2,3) (3,4) (3,5) (3,6) (3,7) (4,8) (4,9) (4,10)